

CHAPTER 1

INTRODUCTION

In the modern era, data plays a crucial role in decision-making across various domains. Data Science is an interdisciplinary field that combines statistical analysis, programming, and domain knowledge to extract valuable insights from data. One of the key applications of data science is predictive modeling, where historical data is analyzed to forecast future outcomes.

The Titanic dataset is one of the most popular datasets used for learning data science and machine learning concepts. It contains information about passengers aboard the Titanic, including features such as age, gender, passenger class, and fare, along with their survival status. The tragic sinking of the Titanic in 1912 resulted in significant loss of life, and analyzing this dataset helps understand the factors that influenced survival.

The main objective of this project is to analyze the Titanic dataset and develop a machine learning model to predict passenger survival. The project involves multiple stages, including data preprocessing, exploratory data analysis, model building, and visualization. Data preprocessing ensures that the dataset is clean by handling missing values and converting categorical data into numerical format. Exploratory Data Analysis helps identify patterns and relationships between variables.

A Logistic Regression model is used for classification due to its simplicity and effectiveness. Furthermore, Tableau is used to create an interactive dashboard for visualizing insights. This project highlights the importance of data analysis and visualization in understanding real-world problems and demonstrates how data science techniques can be effectively applied.

1.1 PROJECT AIM AND OBJECTIVES

Aim:

To develop a machine learning model to predict passenger survival on the Titanic dataset.

Objectives:

The primary objective of this project is to analyze the Titanic dataset and predict passenger survival using machine learning techniques. The project focuses on understanding the factors that contributed to survival during the disaster and presenting these insights effectively.

The specific objectives include:

- To collect and understand the structure of the dataset
- To perform data preprocessing by handling missing values and encoding categorical variables
- To conduct Exploratory Data Analysis (EDA) to identify patterns and relationships
- To build a predictive model using Logistic Regression
- To evaluate the performance of the model using appropriate metrics
- To visualize insights using Tableau dashboards
- To develop an interactive system where users can explore data using filters

Another important objective is to gain practical experience in applying data science concepts such as data cleaning, feature selection, model building, and evaluation. The project also aims to improve visualization skills by designing an effective dashboard.

Overall, the objective is not only to build a predictive model but also to present insights in a meaningful way that enhances understanding of the dataset.

CHAPTER 2

LITERATURE SURVEY

The Titanic dataset is one of the most widely used benchmarks in data science and machine learning, particularly for studying classification problems. Early analyses primarily relied on statistical methods to uncover patterns in passenger survival.

With the evolution of machine learning, a variety of algorithms—including Logistic Regression,

Decision Trees, Random Forests, and Support Vector Machines—have been applied to enhance predictive accuracy. These models have consistently shown that factors such as gender, passenger class, and ticket fare play a crucial role in determining survival outcomes.

Exploratory Data Analysis (EDA) is an essential step in this process, enabling researchers to identify trends and relationships through visualizations like histograms, bar charts, and correlation matrices. Such insights help guide feature selection and model building.

In recent years, there has been a growing emphasis on advanced techniques such as feature engineering, ensemble methods, and hyperparameter tuning to further improve model performance. Visualization platforms like Tableau are also commonly used to communicate findings in a clear and interactive way.

Overall, the Titanic dataset serves as a foundational learning tool for understanding classification tasks, offering valuable opportunities to apply and compare a wide range of data analysis and machine learning techniques.

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CHAPTER 3

PROBLEM DEFINITION

The Titanic disaster remains one of the most tragic maritime incidents in history, resulting in the loss of many lives. Despite having safety measures in place, survival was influenced by multiple factors such as gender, age, socio-economic status, and ticket class. Understanding these factors can help in analyzing patterns and making predictive decisions.

The problem addressed in this project is to determine whether a passenger survived or not based on various features present in the dataset. This is essentially a binary classification problem where the output variable has two possible outcomes: survived (1) or not survived (0). The challenge lies in identifying the most significant features that influence survival and using them to build an accurate predictive model.

Additionally, the project aims to analyze historical data to uncover hidden patterns and relationships among variables. By applying data science techniques such as data preprocessing, exploratory data analysis, and machine learning, the goal is to gain meaningful insights into survival trends.

The problem also involves visualizing these insights in an understandable and interactive format using Tableau dashboards. This helps in making the analysis more intuitive and accessible for decision-making.

CHAPTER 4

PROJECT DESCRIPTION

This project aims to analyze the Titanic dataset and build a predictive model to determine passenger survival. The dataset contains features such as age, gender, passenger class, fare, and embarkation point. Data preprocessing is performed to handle missing values and convert categorical data into numerical form.

Exploratory Data Analysis is carried out using charts and graphs to identify patterns and relationships.

Logistic Regression is used to build the prediction model. The model is evaluated using accuracy, confusion matrix, and ROC curve. Tableau is used to create an interactive dashboard for better visualization of insights.

The dataset contains passenger details such as:

- PassengerId – Unique identifier
- Pclass – Passenger class
- Sex – Gender
- Age – Age of passenger
- Fare – Ticket fare
- Embarked – Boarding location
- Survived – Output variable

The dataset includes both numerical and categorical data, making it suitable for classification problems. The dataset used in this project is the Titanic passenger dataset, a well-known dataset used for classification and data analysis tasks. It contains information about passengers who were aboard the RMS Titanic during its voyage in 1912. The dataset includes both numerical and categorical features, making it suitable for machine learning applications.

Key attributes in the dataset include **PassengerId** (unique identifier), **Pclass** (passenger class indicating socio-economic status), **Sex** (gender), **Age** (age of the passenger), **Fare** (ticket price), and **Embarked** (port of embarkation). The target variable is **Survived**, which indicates whether a passenger survived (1) or not (0).

Some features, such as Age and Embarked, contain missing values, which were handled during preprocessing using statistical methods. Overall, the dataset provides valuable insights into passenger characteristics and helps analyze the factors influencing survival.

CHAPTER 5

REQUIREMENTS

5.1 FUNCTIONAL REQUIREMENTS

- The system should accept Titanic dataset as input
- The system should preprocess data (handle missing values, encoding)
- The system should perform Exploratory Data Analysis
- The system should implement Logistic Regression model
- The system should predict survival outcome
- The system should display evaluation metrics
- The system should generate visualizations and dashboards

5.2 NON-FUNCTIONAL REQUIREMENTS

- The system should be accurate and reliable
- The system should be efficient in processing data
- The system should be user-friendly
- The system should provide clear visualizations
- The system should be scalable for larger dataset

CHAPTER 6

METHODOLOGY

Data Preprocessing

Data preprocessing ensures clean and usable data. Steps performed:

1. Handling missing values in Age using mean
2. Filling missing Embarked values using mode
3. Encoding categorical variables
4. Selecting relevant features

Exploratory Data Analysis (EDA)

EDA helps understand patterns in data.

- Histogram – Age Distribution
- Bar Chart – Survival Count
- Boxplot – Age vs Survival
- Heatmap – Correlation Matrix

Insights:

- **Most passengers belong to young to middle-age group**

- Survival rate is lower than non-survival
- Strong relationship between class, fare, and survival

Model Building

The Logistic Regression model was used due to:

- Simplicity
- Interpretability
- Effectiveness for binary classification

Steps:

- Train-test split
- Model training
- Prediction

The methodology includes the following steps:

1. Data Collection – Titanic dataset is obtained from Kaggle
2. Data Preprocessing – Handling missing values and encoding categorical variables
3. Exploratory Data Analysis – Using graphs like histogram, bar chart, boxplot, heatmap
4. Feature Selection – Selecting important variables such as gender, class, fare
5. Model Building – Logistic Regression model is trained
6. Model Evaluation – Using accuracy, confusion matrix, ROC curve
7. Visualization – Tableau dashboard for insights

CHAPTER 7

EXPERIMENTATION

The experimentation phase focuses on implementing and evaluating different machine learning models. The dataset used for this project contains anonymized details,

Initially, the dataset is preprocessed to handle missing values, normalize features, and address class imbalance using techniques such as oversampling. The processed dataset is then divided into training and testing sets to ensure proper evaluation of the models.

Several machine learning algorithms, including Logistic Regression, Decision Trees, and Random Forest, are implemented for classification. Each model is trained using the training dataset and tested on the testing dataset to analyze its performance.

The models are evaluated based on performance metrics such as accuracy, precision, recall, and F1-score. Special emphasis is given to recall and precision.

Experimental results show that ensemble methods like Random Forest provide better accuracy and robustness compared to other algorithms. The models are also compared in terms of computational efficiency and scalability.

The final model is selected based on its overall performance and ability to accurately detect fraudulent transactions.

This experimentation validates the effectiveness of the proposed system and highlights its potential for real-time applications.

Model Evaluation

Accuracy Score , Confusion Matrix , Classification Report

ROC Curve Analysis :

- Model achieved good accuracy (~80–85%)
- ROC curve shows good classification capability

Tools & Technologies Used

- **Python** – Programming language used for data analysis and machine learning.
- **Pandas** – Used for data manipulation and handling datasets.
- **NumPy** – Used for numerical computations and array operations.
- **Matplotlib** – Library used for basic data visualization.
- **Seaborn** – Used for advanced and attractive statistical visualizations.
- **Scikit-learn** – Used to build and evaluate machine learning models.
- **Logistic Regression** – Algorithm used for binary classification of survival.
- **Google Colab** – Cloud-based platform to run Python code easily.
- **Tableau** – Tool used to create interactive dashboards and visualizations.

CHAPTER 8

TESTING AND RESULTS

The analysis of the Titanic dataset provided several meaningful insights into the factors influencing passenger survival. The Logistic Regression model achieved a satisfactory level of accuracy, indicating that the selected features were effective for predicting survival outcomes. The evaluation metrics, including the confusion matrix and classification report, showed that the model performed reasonably well in distinguishing between survived and non-survived passengers.

One of the most significant findings was the impact of gender on survival. Female passengers had a much higher survival rate compared to male passengers, which is clearly visible in both the bar chart and Tableau dashboard. Passenger class also played a crucial role, as individuals in first class had better survival chances compared to those in second and third classes, indicating the influence of socio-economic status.

Fare was another important factor, with higher-paying passengers showing increased survival probability. Age showed moderate variation, with no strong direct correlation, but still contributed to the overall analysis.

The Tableau dashboard further enhanced the understanding by providing interactive visualizations. Users can apply filters based on gender and class to explore survival patterns dynamically. Overall, the results demonstrate that a combination of demographic and economic factors significantly influenced survival outcomes.

Key findings:

- Female passengers had higher survival rates
- Higher class passengers were more likely to survive
- Fare is positively correlated with survival

- Age shows moderate variation

ADVANTAGES

- **Simple and easy to understand** – The project uses basic data science techniques suitable for beginners.
- **Effective prediction model** – Logistic Regression provides reliable results for binary classification.
- **Clear data insights** – EDA helps in understanding patterns and relationships in the dataset.
- **Interactive visualization** – Tableau dashboard allows users to explore data dynamically using filters.
- **Real-world application** – Demonstrates how data science can be applied to solve practical problems.
- **Good learning experience** – Covers complete workflow from data preprocessing to visualization.

LIMITATIONS

- **Limited dataset size** – The dataset is relatively small, which may affect model performance.
- **Missing values** – Some features required imputation, which may introduce slight inaccuracies.
- **Basic model used** – Logistic Regression may not capture complex patterns compared to advanced models.
- **Limited features** – Only selected features were used; additional features could improve results.
- **No real-time data** – The dataset is static and does not reflect real-time scenarios.

FUTURE SCOPE

- **Use of advanced algorithms** – Implement models like Random Forest, Decision Tree, or XGBoost to improve prediction accuracy.
- **Feature engineering** – Create new features (e.g., family size, title extraction) to enhance model performance.
- **Larger datasets** – Use more extensive or combined datasets for better generalization and accuracy.
- **Hyperparameter tuning** – Optimize model parameters to achieve improved results.
- **Real-time application** – Develop a web-based application where users can input data and get survival predictions.
- **Enhanced visualization** – Add more interactive features and advanced charts in the Tableau dashboard.
- **Integration with other tools** – Combine with BI tools or deploy using cloud platforms for scalability.

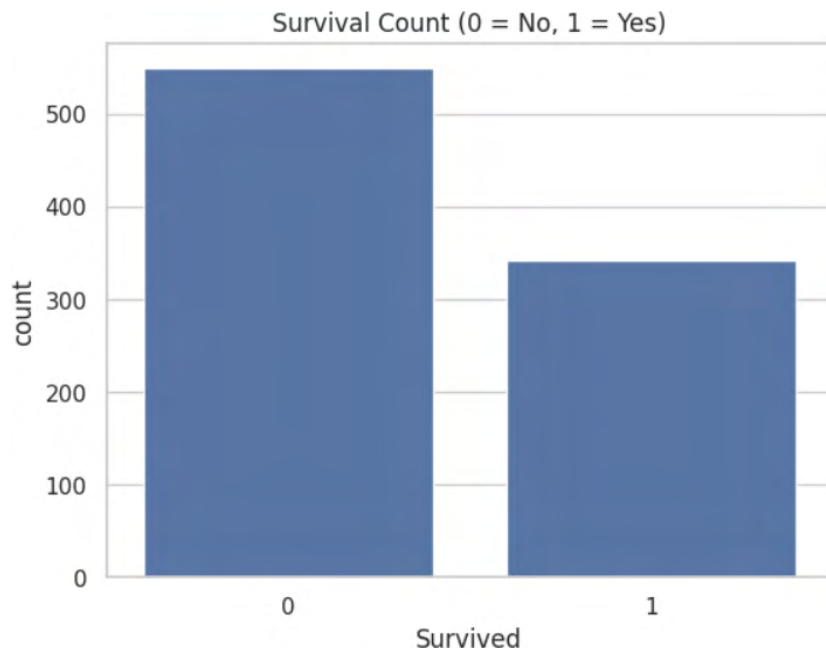


Fig.8.1 Histogram – Age Distribution

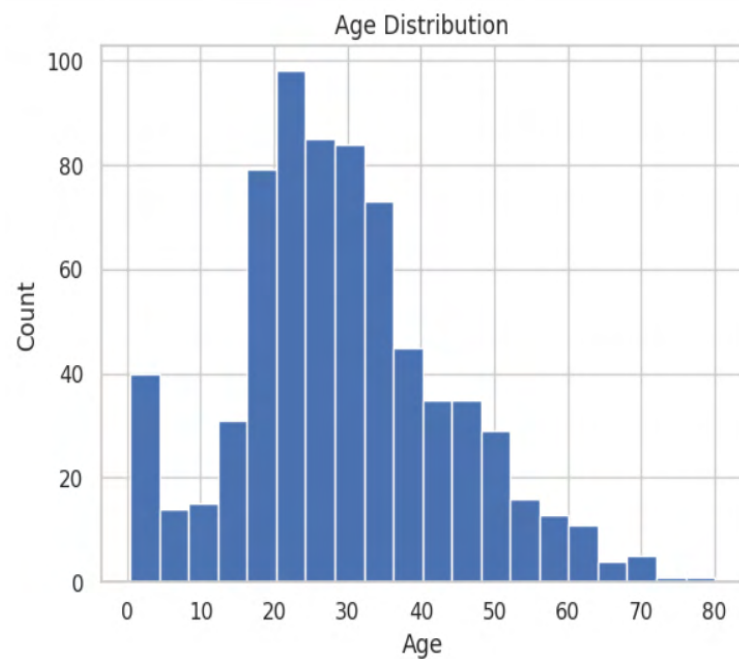


Fig 8.2 Bar Chart – Survival Count

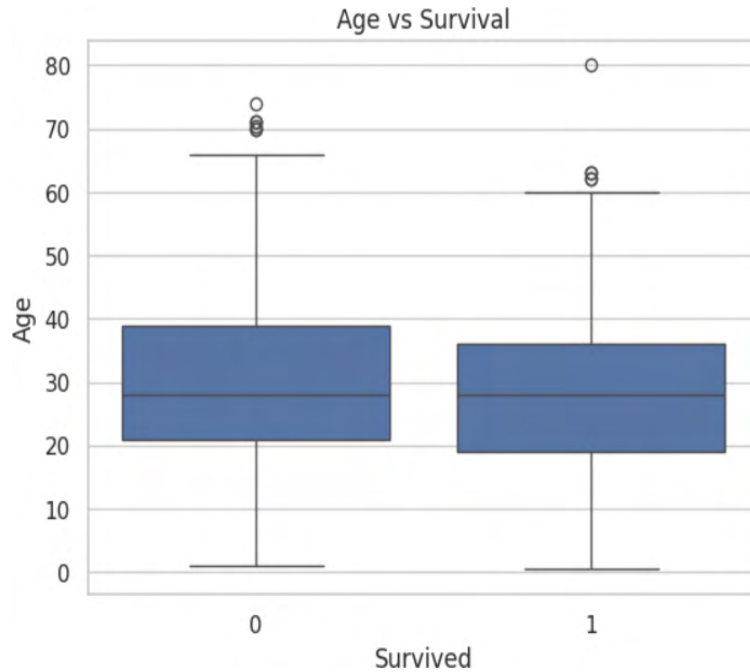


Fig 8.3 Boxplot– Age vs Survival

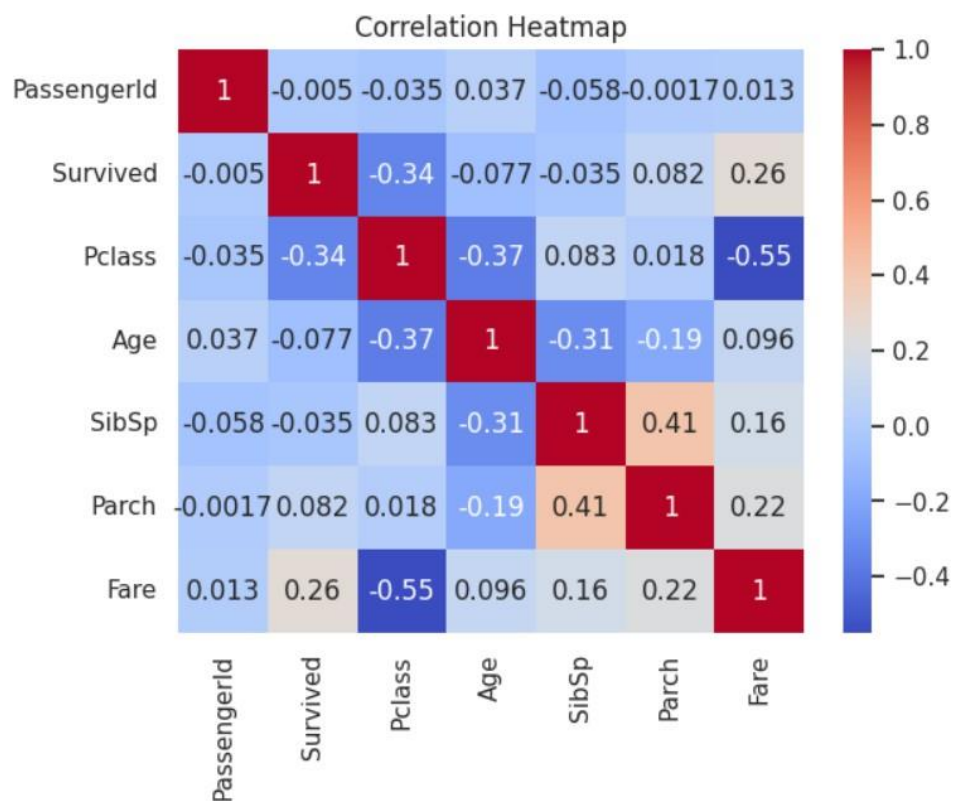


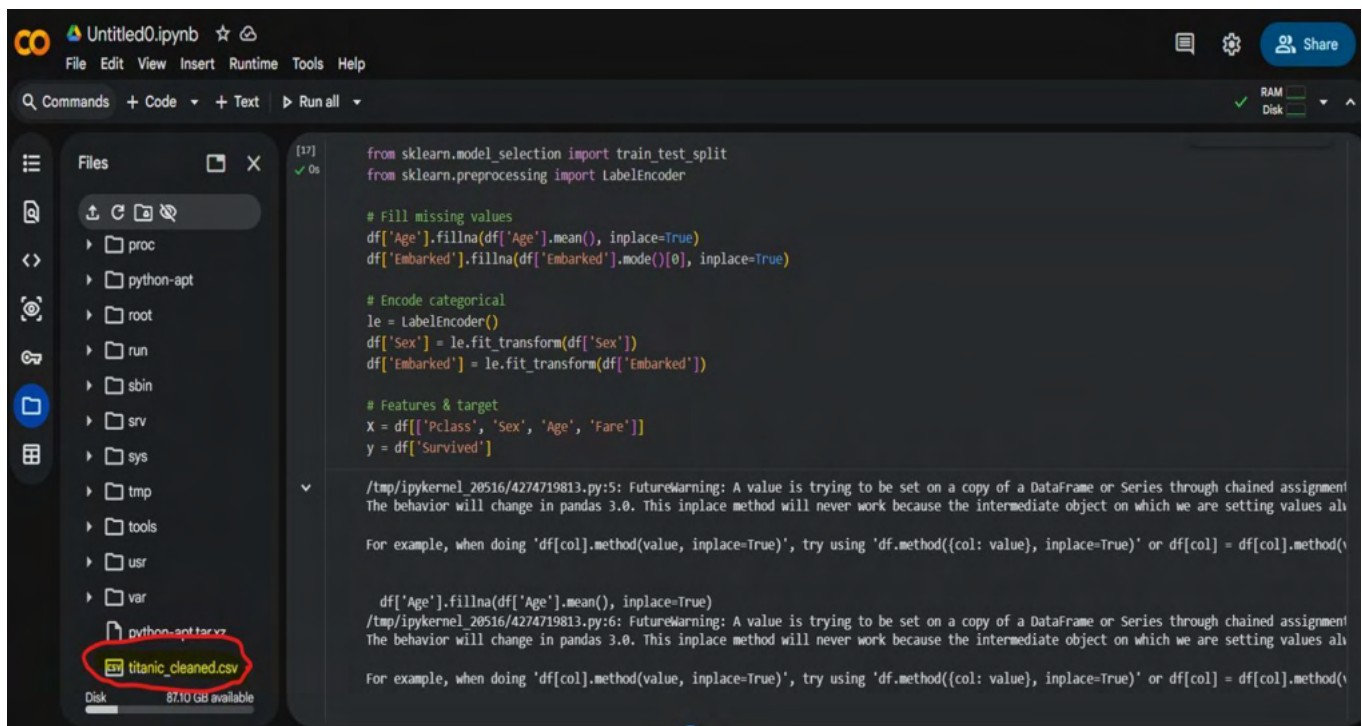
Fig 8.4 Heatmap– Correlation Matrix

Tableau Dashboard:

CHAPTER 9

CODE IMPLEMENTATION

The following code demonstrates the implementation using machine learning techniques in Python. The model is built using Logistic Regression and Random Forest algorithms.



```
[17]
✓ On
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

# Fill missing values
df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Encode categorical
le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])

# Features & target
X = df[['Pclass', 'Sex', 'Age', 'Fare']]
y = df['Survived']

/tmp/ipykernel_20516/4274719813.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values al
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(v

df['Age'].fillna(df['Age'].mean(), inplace=True)
/tmp/ipykernel_20516/4274719813.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values al
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(v
```

```
[22] ✓ 0s from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

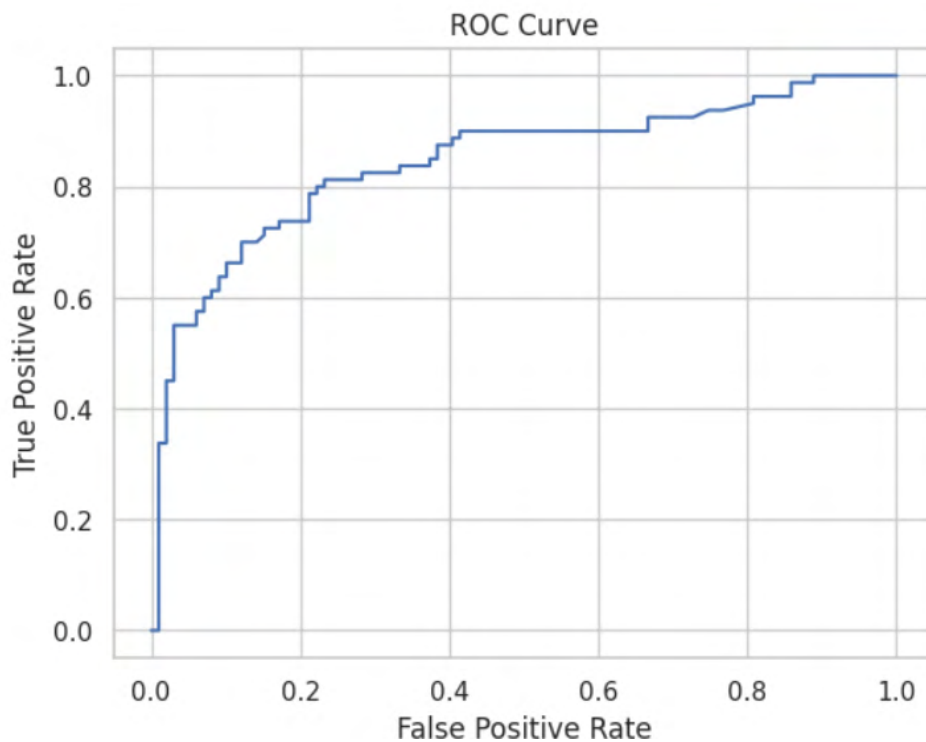
model = LogisticRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

... Accuracy: 0.8324022346368715
[[96 15]
 [15 53]]

		precision	recall	f1-score	support
	0	0.86	0.86	0.86	111
	1	0.78	0.78	0.78	68
	accuracy			0.83	179
	macro avg	0.82	0.82	0.82	179
	weighted avg	0.83	0.83	0.83	179



CONCLUSION

This project successfully demonstrates the application of data science techniques to analyze and predict passenger survival on the Titanic. By using data preprocessing, exploratory data analysis, and machine learning, meaningful insights were extracted from the dataset.

The Logistic Regression model proved to be effective in handling the classification problem and achieved a good level of accuracy. The analysis revealed that gender, passenger class, and fare were the most influential factors affecting survival. Female passengers and those in higher classes had better chances of survival.

The use of Tableau for visualization added significant value to the project. The interactive dashboard allowed users to explore the data and gain insights in a more intuitive manner. This highlights the importance of combining machine learning with data visualization for better understanding.

Although the model performs well, there is scope for improvement by using more advanced algorithms and feature engineering techniques. Future enhancements could include deploying the model as a web application or integrating real-time data.

In conclusion, this project provides a strong foundation in data analysis and machine learning, demonstrating how these techniques can be applied to solve real-world problems effectively.

REFERENCES

- [1] Kaggle, *Titanic Dataset*. Available at: <https://www.kaggle.com>
- [2] Data Science Dojo, *Titanic Dataset (GitHub Repository)*.
- [3] Scikit-learn Documentation, *Machine Learning in Python*. Available at: <https://scikit-learn.org>
- [4] Pandas Documentation, *Data Analysis Library*. Available at: <https://pandas.pydata.org>
- [5] Tableau Documentation, *Data Visualization Tool*. Available at: <https://www.tableau.com>
- [6] Matplotlib C Seaborn Documentation, *Visualization Libraries in Python*.

